

SQL Queries:

- 1) Find the names of artists who have recorded more than one song.
- 2) List the most populous song by an artist.
- 3) List the albums whose downloads count have exceeded 1000.
- 4) List all the romantic albums recorded by "Kumar Sanu".
- 5) List all the composers who have been born after 1980, and whose albums have been downloaded more than 500 times.
- 6) List all the songs produced by male producers whose sales have crossed 450.
- 7) List all the patriotic songs recorded by multiple artists produced after 1990.
- 8) Delete all the songs by "Sonu Nigam", update the database with songs of "Atif Aslam".
- 9) Find out the names of all composers aging below 25 who have composed atmost 2 songs.
- 10) List all the songs produced under the banner "Eros", also sung by "Akriti Kakkar".
- 11) List all the songs sung by "Atif Aslam" under the banner "T-Music" after 2005.
- 12) List the total downloads' count.

i) Table Creation

```
create table ARTIST
(
  Artist_ID varchar(6) primary key check(Artist_ID like 'Ar%'),
  Ar_Name varchar(20) not null,
  Ar_Sex varchar(10) not null,
  Ar_DOB date not null,
  Ar_Age number(3) not null
);
```

```
create table ALBUM
(
  Album_ID varchar(6) primary key check(Album_ID like 'A%'),
  Genre varchar(20) not null,
  R_Date date,
```

```
NO_Downloads number(6) not null,  
NO_Sales number(6) not null  
);
```

```
create table COMPOSER  
(  
  Composer_ID varchar(6) primary key check(Composer_ID like 'C%'),  
  C_Name varchar(20) not null,  
  C_Sex varchar(10) not null,  
  C_DOB date not null,  
  C_Age number(3) not null  
);
```

```
create table LYRICIST  
(  
  Lyricist_ID varchar(6) primary key check(Lyricist_ID like 'L%'),  
  L_Name varchar(20) not null,  
  L_Sex varchar(10) not null,  
  L_DOB date not null,  
  L_Age number(3) not null  
);
```

```
create table PRODUCER
(
  P_ID varchar(6) primary key check(P_ID like 'P%'),
  P_Name varchar(20) not null,
  P_Sex varchar(10) not null,
  P_DOB date not null,
  P_Age number(3) not null
);
```

```
create table SONG
(
  S_ID varchar(6) primary key check(S_ID like 'S%'),
  S_Name varchar(30) not null,
  Artist_ID varchar(6) not null,
  Album_ID varchar(6) not null,
  Composer_ID varchar(6) not null,
  Lyricist_ID varchar(6) not null,
  P_ID varchar(6) not null,
  S_Sales number(6) not null,
  foreign key(Artist_ID) references ARTIST(Artist_ID) on delete cascade,
  foreign key(Composer_ID) references COMPOSER(Composer_ID) on delete cascade,
  foreign key(Lyricist_ID) references LYRICIST(Lyricist_ID) on delete cascade,
  foreign key(P_ID) references PRODUCER(P_ID) on delete cascade,
  foreign key(Album_ID) references ALBUM(Album_ID) on delete cascade
);
```

ii) Insertion

In this section, we are displaying each and every view after it is inserted with all the desired values

```
Command Prompt - sqplus
SQL> Select * from ARTIST;

ARTIST_AR_NAME      AR_SEX  AR_DOB      AR_AGE
-----
Ar001  Akriti Kakkar    Female    07-AUG-86      33
Ar002  Arijit Singh     Male     25-APR-87      33
Ar003  Kunal Sanu       Male     20-OCT-57      62
Ar004  Atif Aslam       Male     12-MAR-83      37
Ar005  Sonu Nigam       Male     30-JUL-73      46

SQL> Select * from ALBUM;

ALBUM_GENRE      R_DATE      NO_DOWNLOADS  NO_SALES
-----
A0001  Romantic      24-NOV-11      1000      250
A0002  Comedy        12-JUN-09      1150      175
A0003  Horror        14-SEP-04      1340      500
A0004  Tragic        07-NOV-01      600      325
A0005  Drama         09-JAN-15      800      600
A0006  Patriotic     03-AUG-10      300      400

6 rows selected.

SQL> Select * from COMPOSER;

COMPOS_C_NAME      C_SEX  C_DOB      C_AGE
-----
C001  Bagchi       Male     13-JUN-97      23
C002  Rahman      Male     12-OCT-89      30
C003  Clerk       Male     14-DEC-97      24

SQL> Select * from LYRICIST;

LYRICI_L_NAME      L_SEX  L_DOB      L_AGE
-----
L0001  Javed Akhtar  Male     14-JUN-78      42
L0002  Begum Akhtar  Female   04-NOV-64      56
L0003  Jenny Rodriques Mon-Binary 01-DEC-93      26
L0004  Milind Soman  Queen    12-MAY-81      39

SQL> Select * from PRODUCER;

P_ID  P_NAME      P_SEX  P_DOB      P_AGE
-----
P001  Eros        N/A    12-SEP-59      60
P002  Ratan Shah  Male   12-MAR-66      54
P003  T-Music     N/A    12-MAY-75      45
P004  Vijaylakshmi Female  13-AUG-81      38
P005  Surtal      N/A    14-APR-40      80
```

```
Command Prompt - sqplus
SQL> Select * from SONG;

S_ID  S_NAME      ARTIST  ALBUM  COMPOS  LYRICI  P_ID
-----
S001  Abhi Abhi    Ar001  A0001  C001   L0001  P001
S002  Baarish     Ar002  A0002  C002   L0004  P003
S003  Dheere Dheere Ar003  A0003  C001   L0001  P002
S004  Bas Ek Sanam Ar003  A0003  C001   L0001  P001
S005  Tu Jaane Na Ar004  A0004  C003   L0002  P005
S006  Suraj       Ar005  A0006  C003   L0003  P004
S007  Main Agar   Ar005  A0006  C003   L0004  P004
S008  Saathiya    Ar001  A0001  C002   L0004  P001
S009  Dil Diyaan  Ar004  A0005  C002   L0003  P004
S010  Mast Nagan   Ar002  A0002  C002   L0002  P003
```

iii) Queries

q) Find the names of artists who have recorded more than one song

```
SQL> select * from ARTIST
  2  where Artist_ID in (
  3  select Artist_ID from SONG
  4  group by Artist_ID
  5  having count(*)>1
  6  );
```

ARTIST	AR_NAME	AR_SEX	AR_DOB	AR_AGE
Ar001	Akriti Kakkar	Female	07-AUG-86	33
Ar002	Arijit Singh	Male	25-APR-87	33
Ar003	Kumar Sanu	Male	20-OCT-57	62
Ar004	Atif Aslam	Male	12-MAR-83	37
Ar005	Sonu Nigam	Male	30-JUL-73	46

q) List the most populous sung by an artist

```
SQL> select * from SONG
  2  where S_Sales = (
  3  select max(S_Sales) from SONG
  4  );
```

S_ID	S_NAME	ARTIST	ALBUM_	COMPOS	LYRICI	P_ID
S009	Dil Diyaan	Ar004	A0005	C002	L0003	P004
	600					

q) List the albums whose downloads count have exceeded 1000

```
SQL> select * from ALBUM
  2  where NO_Downloads > 1000;
```

ALBUM_	GENRE	R_DATE	NO_DOWNLOADS	NO_SALES
A0002	Comedy	12-JUN-09	1150	175
A0003	Horror	14-SEP-04	1340	500

q) List all the composers who have been born after 1980, and whose albums have been downloaded more than 500 times

```
SQL> select * from COMPOSER
  2  where Composer_ID in (
  3      select Composer_ID from COMPOSER
  4      natural join SONG
  5      natural join ALBUM
  6      where C_DOB > '31-DEC-1980' and NO_Downloads > 500
  7  );
```

COMPOS	C_NAME	C_SEX	C_DOB	C_AGE
C001	Bagchi	Male	13-JUN-97	23
C002	Rahman	Male	12-OCT-89	30
C003	Clerk	Male	14-DEC-97	24

q) List all the songs produced by male producers whose sales have crossed 450

```
SQL> select * from PRODUCER
  2  where P_Sex = 'Male' and P_ID in (
  3      select P_ID from SONG
  4      where S_Sales > 450
  5  );
```

P_ID	P_NAME	P_SEX	P_DOB	P_AGE
P002	Ratan Shah	Male	12-MAR-66	54

q) List all the patriotic songs recorded by multiple artists produced after 1990

```
SQL> select * from SONG
  2  where Album_ID in (
  3      select Album_ID from ALBUM
  4      where Genre = 'Patriotic'
  5  );
```

S_ID	S_NAME	ARTIST	ALBUM_	COMPOS	LYRICI	P_ID
S006	Suraj 400	Ar005	A0006	C003	L0003	P004
S007	Main Agar 400	Ar005	A0006	C003	L0004	P004

q) Delete all the songs by "Sonu Nigam" , update the database with songs of Atif Aslam

```
SQL> delete from SONG
2 where Artist_ID = 'Ar005';

2 rows deleted.

SQL> update SONG
2 set Artist_ID = 'Ar004';

8 rows updated.

SQL> SELECT *FROM SONG;

S_ID  S_NAME          ARTIST ALBUM_ COMPOS LYRICI P_ID
-----
S_SALES
-----
S001  Abhi Abhi       Ar-004 A0001 C001  L0001 P001
250
S002  Baarish        Ar-004 A0002 C002  L0004 P003
175
S003  Dheere Dheere    Ar-004 A0003 C001  L0001 P002
500
S_ID  S_NAME          ARTIST ALBUM_ COMPOS LYRICI P_ID
-----
S_SALES
-----
S004  Bas Ek Sanam   Ar-004 A0003 C001  L0001 P001
250
S005  Tu Jaane Na    Ar-004 A0004 C003  L0002 P005
325
S008  Saathiya       Ar-004 A0001 C002  L0004 P001
250
S_ID  S_NAME          ARTIST ALBUM_ COMPOS LYRICI P_ID
-----
S_SALES
-----
S009  Dil Diyaan    Ar-004 A0005 C002  L0003 P004
600
S010  Mast Hagan     Ar-004 A0002 C002  L0002 P003
175
```

q) Find out the names of all composers aging below 25 who have composed at most 2 songs

```
SQL> select distinct(C_Name) from SONG
2 natural join COMPOSER
3 where C_Age < 25;

C_NAME
-----
Clerk
Bagchi
```


q) List all the songs produced under the banner "Eros", also sung by "Akriti Kakkar"

```
SQL> select * from SONG
2  where P_ID = 'P001' and Artist_ID = 'Ar001';
```

S_ID	S_NAME	ARTIST	ALBUM_	COMPOS	LYRICI	P_ID
S001	Abhi Abhi 250	Ar001	A0001	C001	L0001	P001
S008	Saathiya 250	Ar001	A0001	C002	L0004	P001

q) List all the songs sung by "Atif Aslam" under the banner "T-Music" after 2005

```
SQL> select * from SONG
2  where Artist_ID = 'Ar004' and Album_ID in (
3      select Album_ID from ALBUM
4      where R_Date > '31-DEC-2005'
5  );
```

S_ID	S_NAME	ARTIST	ALBUM_	COMPOS	LYRICI	P_ID
S009	Dil Diyaan 600	Ar004	A0005	C002	L0003	P004

q) List the total downloads' count

```
SQL> select sum(NO_Downloads) from ALBUM;
```

SUM(NO_DOWNLOADS)
5190

Normalisation

Normalization is the process of organizing the data in the database. **Normalization** is used to minimize the redundancy from a relation or set of relations. It is also used to eliminate the undesirable characteristics like Insertion, Update and Deletion Anomalies

Here are the most commonly used normal forms:

- First normal form(1NF)
- Second normal form(2NF)
- Third normal form(3NF)
- Boyce & Codd normal form (BCNF)

Our database is normalised and it is in BCNF(Boyce Codd's Normal Form). Reasons are:-

1)Atomicity preserved, satisfying **1NF**

2)No non-prime attributes is dependent on the proper subset of the candidate key, satisfying **2NF**

3)Removal of transitive functional dependency and ensuring each of the attributes is directly dependent on the super key/candidate key, satisfying **3NF**

4)Since, all the attributes from each and every table are **strictly** dependent on the super key, thus database is in BCNF(**Stricter** form of 3NF)

Relational model:

